DO NOW — RECALL

Waterfall, Agile and Extreme Programming

Name: _____ Class: _____ Date: _____

1. Which OCR pseudocode function reads the next line of a text file?

2. What is a function parameter?

3. Why use boundary-value testing?

4. If requirements frequently change, what is the risk of freezing all requirements before implementation?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

readLine() on an opened file object.

2. Model answer

A named input received by a subroutine.

3. Model answer

Defects often occur at the limits of valid ranges.

4. Model answer

Large amounts of design/code may need reworking when changes emerge.

Vocabulary — know these terms

Waterfall — Primarily sequential software-development life cycle.

Agile — Iterative approach using feedback and incremental releases.

XP — Extreme Programming, an agile methodology with disciplined engineering practices.

sprint — Short time-boxed development iteration.

test-driven development — Write an automated test before writing the production behaviour.

NEW LEARNING

Read, annotate and explain

Waterfall is sequential

Agree requirements, then design, implementation, testing and deployment. Revisiting an earlier stage can be costly.

Agile delivers incrementally

Each short iteration produces working software and seeks feedback, so priorities can change between iterations.

Extreme Programming (XP)

XP is an agile methodology emphasising practices such as test-driven development, pair programming, refactoring and continuous integration.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A hospital requests software with a carefully agreed specification and formal approvals. Why might Waterfall suit it?

Predict first

Requirements and approval milestones can be documented before later stages proceed.

Model and explain

Waterfall offers clear documentation and control, but late changes can require expensive rework.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State a defining characteristic of Waterfall, Agile and XP.

Question 2

A mobile app company receives new customer requirements each fortnight. Recommend a methodology and explain two advantages and one disadvantage.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain how test-driven development and continuous integration in XP could reduce defects, and why neither guarantees fault-free software.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Agile means software is never planned or tested.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.